Assembly, formation mechanism, and enhanced gas-sensing properties of porous and hierarchical SnO_2 hollow nanostructures

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Hierarchical and hollow SnS_2 nanostructures as precursors were fabricated via a surfactant-assisted assembly process using sodium dodecyl sulfate as soft templates. The as-prepared SnS_2 nanostructures were further oxidized to form porous SnO_2 conversion for investigating their gas-sensing properties in drug-precursor detection. On the basis of a series of time- and ratio-dependent reactions, a formation mechanism of the special nanostructures and factors influencing morphology and structure were determined. Gas-sensing measurements revealed that the porous and hierarchical SnO_2 hollow nanostructures were sensitive to drug precursors, indicating promising applications in environmental monitoring and public safety investigation. In addition, we found that the assembled SnO_2 nanomaterials possessed significantly enhanced gas-sensing properties compared with unassembled SnO_2 with a solid interior.

I. INTRODUCTION

Hierarchical architectures, especially micro/nanoscale structures, have attracted extensive attention due to their special properties that enable them to be applied in optics, electronics, magnetics, biomimetics, and catalysis.[1–4] As the performance of such applications are determined by the morphology and structure of materials, during the past decade, effort has focused on the preparation of well-defined hierarchical nanoarchitectures and more complex nanostructures via various solution- or vapor-phased approaches.[5–7] Among them, the surfactant-assisted assembly process is of particular interest because of its advantages, for example, low cost, large yield, and excellent repeatability. Many surfactants, such as cetyltrimethyl ammonium bromide (CTAB), poly 4-nonylphenyl 3-sulfopropyl ether potassium salt (PENS), sodium dodecyl benzenesulfonate (SDBS), and polyvinyl pyrrolidine (PVP), have been developed as soft templates for the assembly of hierarchical nanostructures, for example, ZnO, $Ni(OH)_2$, CdS, and MnO_2.[8–10] In spite of these significant achievements, developing novel surfactant-assisted processes to assemble hierarchical nanoarchitectures for applications still remains a great challenge.

In recent years, sodium dodecyl sulfate (SDS) has received increasing interest due to its special molecular structure with a polar group and a tail-like nonpolar group. Lee et al. reported the assembly of organic materials from nanoparticles to nanorods and nanowires by using SDS as the template.[11] More recently, Edler's group revealed that polymers could interact with the SDS/CTAB mixture, thereby enabling it to fabricate more complex structures.[12] Furthermore, as SDS molecules possess strong coordination ability toward metal ions, it is indicated that SDS would be a promising candidate for assembling novel inorganic hierarchical nanostructures.

SnS_2, as one of the IV-VI semiconductors, possesses interesting photoelectric properties with potential applications in optoelectronic devices.[13,14] More important, it is the ideal precursor for fabricating SnO_2 via an oxidizing process. SnO_2 as an n-type semiconductor with a wide band gap ($E_g = 3.6$ eV at 300 K), it has been widely used in gas sensors, lithium-ion batteries, and solar cells, and it has been considered as one of the most important semiconductors in the electronics industry.[15–18] It is known that surface area plays a key role in influencing the performance of electronic devices based on contact reactions. As for gas sensors, a large surface area of the sensing material is crucial for excellent gas-sensing performance.[19–21] For example, Martinez et al. reported that a multiscale porous structure could effectively increase the active surface area of gas-sensing SnO_2 materials,

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thereby leading to improved sensitivity.[22] Accordingly, design and preparation of SnO_2 structures with a large surface area is of great value for performance enhancement for SnO_2-based electronic devices.

Herein, we present an SDS-assisted assembly process for the fabrication of hierarchical and hollow SnS_2 nanoarchitectures. The as-assembled SnS_2 nanostructures as precursors were further oxidized to prepare porous SnO_2 conversion for gas-sensing properties investigation, as shown in Scheme 1. In gas-sensing measurements, drug precursors, including acetone, chloroform, and ethyl ether, which not only do serious harm to lungs and nerves but also can be used to synthesize drugs as raw materials or additives,[23,24] were used as target gases. The results show that a gas sensor based on porous and hierarchical SnO_2 hollow nanoarchitectures exhibits sensitivity to drug precursors, indicating its potential applications in environmental monitoring and public safety investigation. The mechanism of the SDS-assisted assembly process and factors influencing the morphology and structure of products were also investigated.

II. EXPERIMENTAL SECTION

A. Assembly and oxidization of SnS_2 precursors

All chemicals were analytical grade and used without further purification as purchased from Shanghai Chemical Reagents Company (Shanghai, China). In a typical procedure, 10 mmol SDS was dissolved into a certain amount of deionized water to form a homogeneous solution (30 mL) by constant stirring. Next, 1 mmol $SnCl_4{\cdot}5H_2O$ and 2 mmol L-cysteine were added to the as-obtained solution following by alternate stirring and ultrasonication. The mixture was transferred into a Teflon-lined stainless steel autoclave with a capacity of 40 mL. The autoclave was sealed and kept in an oven at a constant temperature of 160 °C for 16 h. Subsequently, the autoclave was taken out and allowed to cool to room temperature naturally. The precipitate was collected by centrifugation and alternately washed several times with ethanol and deionized water, and dried under vacuum. To prepare porous SnO_2 structures, the as-obtained SnS_2 products were further oxidized by annealing in a furnace at 550 °C for 2 h in air.

B. Characterization

The x-ray diffraction (XRD) pattern of the products was recorded on a Philips X' Pert Pro x-ray diffractometer (Almelo, The Netherlands) equipped with the monochromatic high-intensity Cu Kα radiation (1.54178 Å). The energy-dispersive x-ray diffraction (EDXRD) pattern was obtained by using an EDXRD spectrometer (Beijing, China) consisting of a polychromatic source of x-rays, collimators, and an Amptek XR-100T-CdTe detector. The central angle 2θ at which the diffracted beam was detected was fixed at 16°. The morphology and structure of the products were observed by an FEI Sirion 200 field-emission scanning electronic microscopy (FESEM; Eindhoven, The Netherlands), and a Hitachi H-800 transmission electron microscope (Tokyo, Japan) at an accelerating voltage of 200 kV. High-resolution TEM (HRTEM) images, selective area electron diffraction (SAED) pattern, and energy dispersive x-ray spectroscopy (EDX) spectrum were obtained on a JEOL-2010 transmission electron microscope (Tokyo, Japan) equipped with an Oxford windowless Si (Li) detector. Fourier transform infrared (FTIR) absorption spectra were recorded on a Shimadzu IR-440 spectrometer (Kyoto, Japan) from 4000 to 500 cm^{-1}. The Brunauer-Emmett-Teller (BET) surface area of the as-obtained products was measured on a Coulter Omnisorp 100CX instrument (Los Angeles, CA) by using N_2 adsorption and desorption.

C. Fabrication of gas sensor and gas-sensing measurements

There are two steps for the measurement of gas-sensing performance: (i) the fabrication of gas sensors and (ii) gas-sensing detection of target gases. The structure of gas sensor has been presented previously.[20] First, the as-prepared SnO_2 nanoarchitectures were dispersed in absolute alcohol and coated onto the surface of the Al_2O_3 substrate ($5.0 \times 2.0 \times 0.5$ mm^3). The coated substrate was covered with gold electrodes and an RuO_2 layer as the heater, following by a drying process in an oven. Next, gas-sensing detection was done by a computer-controlled gas-detecting system in which a Keithley 6487 picoammeter/voltage sourcemeter (Cleveland, OH) was used as both current recorder and power source. All gas-sensing measurements were carried out at the same working temperature of 200 °C. Vapors of acetone, chloroform, and ethyl ether which belong to drug precursors, at various concentrations ranging from 50 to 500 ppm, were used as target gases.

The sensitivity (S) of the gas sensor was defined as the following equation:

$$S = \frac{R_{air}}{R_{gas}} = \left(\frac{I_{gas}}{I_{air}}\right)_V \quad ,$$

where R_{air} is the resistance in dry air and R_{gas} is that in the mixture of dry air and target gas. According to Ohm's saw, in which current is inversely proportional to resistance under constant voltage, S can also be represented by the current in dry air (I_{air}) and in a gas mixture (I_{gas}) under a constant measuring voltage.

III. RESULTS AND DISCUSSION

A. Morphology, structure, and composition of precursors

Typical XRD and EDXRD patterns of the products prepared by SDS-assisted assembly in a molar ratio

(Sn^{4+}:L-cysteine:SDS) of 1:2:10 at 160 °C for 16 h are shown in Figs. 1(a) and 1(b), respectively. In Fig. 1(a), all diffraction peaks can be indexed to the hexagonal SnS_2 structure with lattice constants of $a = 3.648$ Å and $c = 5.899$ Å, which are in good agreement with literature values (JCPDS Card No. 23-0677). The results obtained above are further confirmed by EDXRD pattern [Fig. 1(b)]. Compared with the standard (angle-dispersive) XRD technique in which a monochromatic beam and a detector that scans through a range of angles are used, the polychromatic x-ray beam used in EDXRD allows an energy-discriminating detector to be fixed at a given angle.[25] By means of Bragg's law ($n\lambda = 2d\sin\theta$), the diffracted beams from crystalline structures with different d spacings can be distinguished by values of λ under a fixed θ in EDXRD, rather than by the values of θ in standard XRD (using fixed λ).[26] In Fig. 1(b), all the calculated d spacings corresponding to different peaks perfectly match with the literature values (JCPDS Card No. 23-0677), further confirming the hexagonal SnS_2 structure of the as-obtained samples. In addition, no peaks from impurities in either the XRD or the EDXRD patterns were detected (the signals of Cu in the EDXRD pattern came from the target), indicating a high purity of the as-assembled products.

Figure 2 shows the morphology of as-prepared SnS_2 samples. Seen from the FESEM image [Fig. 2(a)], a well-defined hierarchical nanostructure which consists of numerous radial nanoflakes is clearly exhibited. The diameter of each spherical architecture is 2 μm, and the thickness of the nanoflakes is ~10 nm. In addition, the high-magnification FESEM image [Fig. 2(b)] reveals that the SnS_2 nanostructures are hollow, which can be further confirmed by TEM image, as shown in Fig. 2(c). As the hollow structure is not obvious, two aspects are proposed in explanation. Firstly, SDS was used as a soft template which could aggregate to form spherical vesicles in hydrothermal conditions.[27,28] In our investigation, hierarchical spheres consist of numerous nanoflakes. Both the large thickness of spheres and the density of the nanoflakes indicate a hollow interior. Secondly, during the formation process, nanoflakes grow along both the external and internal surfaces, resulting in a decrease in the inner diameter and the complication of the interior. In Fig. 2(d), the well-arranged crystalline lattice fringes in

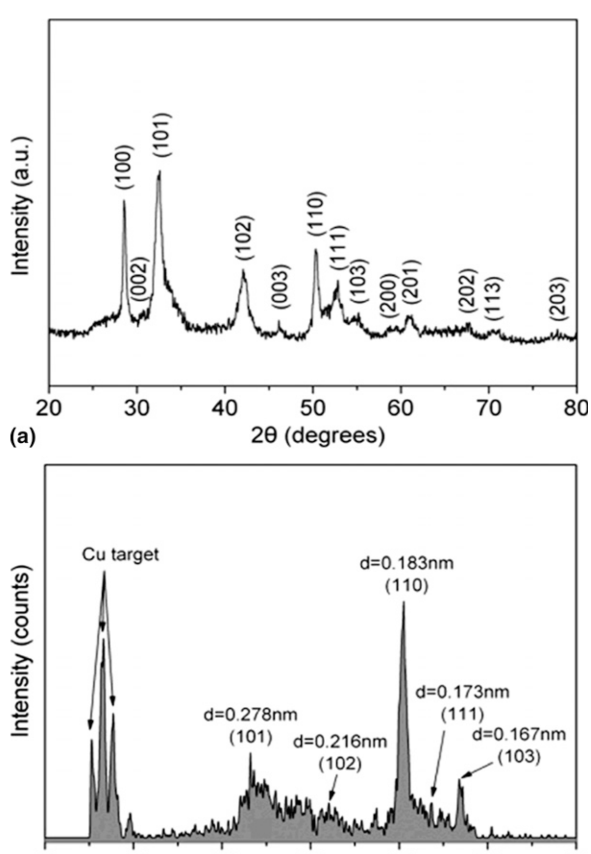


FIG. 1. (a) XRD and (b) EDXRD patterns of the products prepared by SDS-assisted assembly in a molar ratio (Sn^{4+}:L-cysteine:SDS) of 1:2:10 at 160 °C for 16 h.

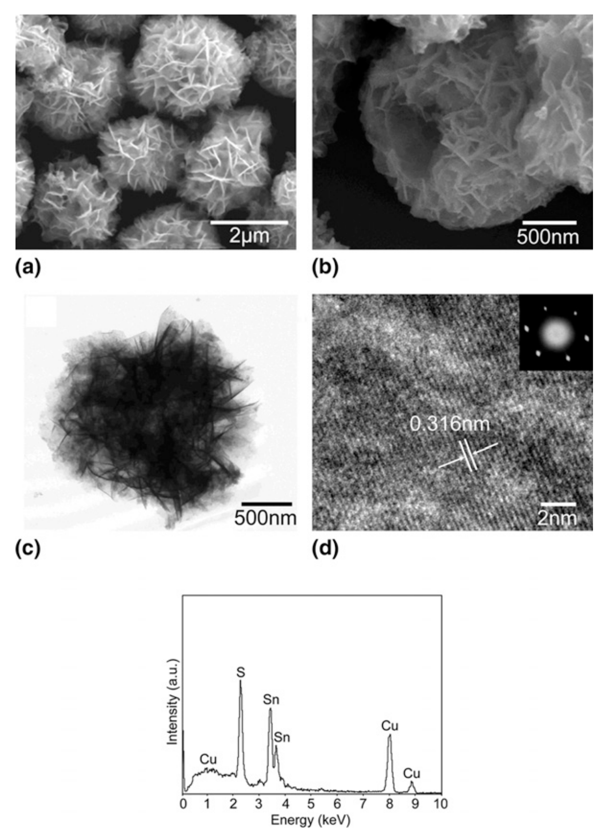


FIG. 2. Typical morphology of the as-assembled SnS_2 nanoarchitecture: (a) low magnification and (b) high magnification FESEM images; (c) TEM photograph; (d) lattice resolved HRTEM images with an inserted corresponding SAED pattern. The composition of samples is presented in (e) EDX spectrum.

the HRTEM image and discrete spots in the SAED pattern indicate a well-developed crystallinity and a single-crystal morphology of the nanoflakes. The interplanar spacing of the planes is approximately 0.316 nm, which is consistent with the *d* spacing of the (100) lattice planes of hexagonal SnS_2. Furthermore, the EDX spectrum [Fig. 2(e)] demonstrates that the as-assembled architectures are composed of Sn and S elements in a molar ratio (Sn:S) of 1:1.989 without impurities, which is in agreement with the stoichiometric ratio of SnS_2.

B. Assembly process and influencing factors

As to the assembly process illustrated in Scheme 1, SDS plays a key role during the formation of hierarchical SnS_2 nanoarchitectures with hollow structure. It is significant to investigate the SDS-assisted assembly process for understanding the growth mechanism of the special SnS_2 structure, controlling the morphology and structure of products, as well as exploring the potential applications of the novel synthetic approach. A series of reactions with different synthetic times ranging from 2 to 16 h were carried out at 160 °C. The FESEM images of products obtained at different assembly stages are shown in Fig. 3. Seen from Fig. 3(a), only nanoparticles are formed at the initial stage. During this period, the spherical templates constructed by SDS molecules have not been achieved yet. Through coordination between SDS and Sn^{4+} ions, aggregation among SnS_2 nuclei is effectively prevented, resulting in the formation of SnS_2 nanoparticles rather than nanoarchitectures or their bulk counterparts. Figure 4 shows FTIR spectra of pure SDS and the products obtained at the initial period of assembly [at 160 °C for 2 h, Fig. 3(a)]. The bands located at 2916 and 2845 cm^{-1} can be assigned to the asymmetric and symmetric stretching vibrations of C–H in methylene chains.[29] The bands at 1160 and 1023 cm^{-1} are attributed to the S–O and S=O stretching modes,[30] which are characteristic of SDS in reagents in our study. As compared with pure SDS, these bands in the FTIR spectrum of initial products indicate the presence of SDS in samples obtained at the initial synthetic stage. Based on the combination of FTIR analysis and the strong coordination ability of SDS toward metal ions, we deduce the coordination between the SDS and the Sn^{4+} ions, and therefore the oriented growth of the SnS_2 nuclei.

As the reaction time continues, spherical architectures with a rough surface are fabricated, as shown in Fig. 3(b). The formation of hierarchical nanoarchitectures develops from the fabrication of spherical parent structures to the secondary growth of nanoflakes, but not from the growth of individual nanoflakes to a complex of nanoflakes. This indicates an enhanced mechanical stability of the hierarchical products in our study. Further increasing the reaction time, more and more sulphur sources are gradually released from L-cystine, resulting in the growth of nanoflakes under hydrothermal conditions, as shown in

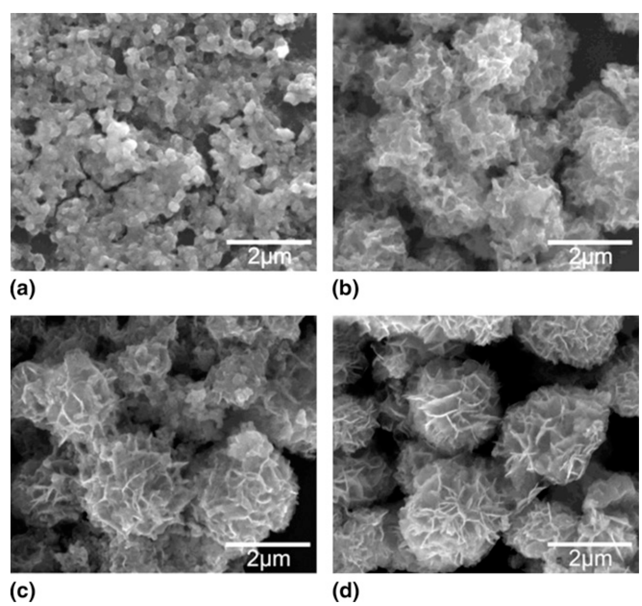
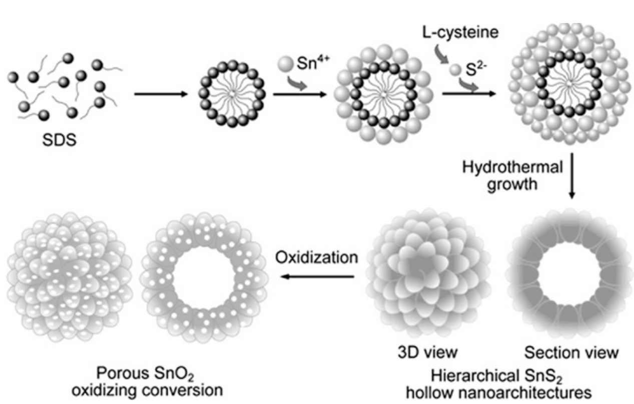


FIG. 3. FESEM images of the samples obtained at different assembly stages at a determined reaction temperature of 160 °C: (a) 2 h, (b) 4 h, (c) 8 h, and (d) 16 h.



SCHEME 1. Schematic illustration of SDS-assisted assembly of hierarchical SnS_2 hollow nanoarchitecture and subsequent oxidizing process.

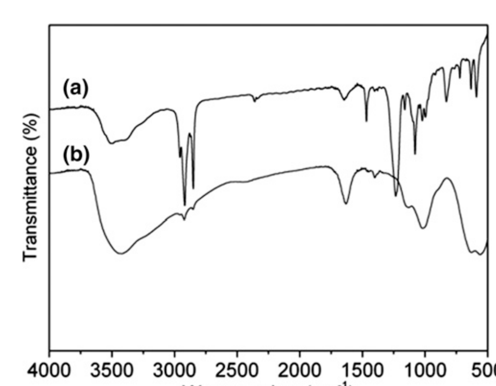


FIG. 4. FTIR spectra of (a) pure SDS and (b) products synthesized at 160 °C for 2 h.

Fig. 3(c). Moreover, as crystallinity continues in this period, coordination between SDS and Sn^{4+} ions gradually decomposes. As the time increases to 16 h, a well-defined hierarchical hollow nanoarchitecture is formed, as shown in Figs. 2 and 3(d).

In order to further demonstrate the role of SDS in the assembly process, and to investigate the potential factors which are significant to influence the morphology and structure of products, various reactions under conditions of different amounts of SDS and a series of molar ratios (Sn^{4+}:L-cystine) were also performed. Figure 5 shows the FESEM images of samples prepared at 160 °C for 16 h under different amounts of SDS. As compared with the final products fabricated under 10 mmol SDS, only thick and large nanosheets can be obtained with less SDS (5 mmol), as shown in Fig. 5(a), which is similar to the reported morphology without SDS-assisted assembly.[31] As illustrated in the inserted schematic diagram, the SDS spherical soft templates cannot be achieved completely in this condition, leading to the formation of numerous fragment-like SnS_2 architectures. In contrast, while SDS is increased to 15 mmol, irregular nanoarchitectures are formed, as displayed in Fig. 5(b). At the other extreme, for excessive SDS, aggregation among SDS templates deteriorates, resulting in the breakage of regular spherical structures. Hence, an optimal amount of SDS in the assembly process is necessary to get a well-defined SnS_2 hierarchical nanostructure, which is 10 mmol in our study, and it would also be applicable for fabricating other hierarchical nanostructures through SDS-assisted assembly process.

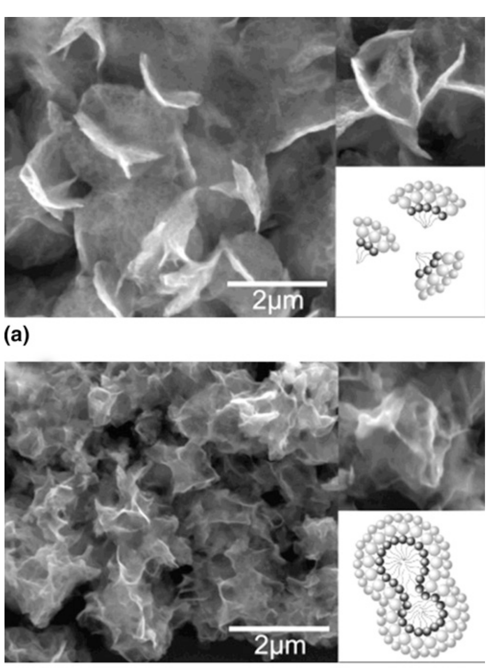


FIG. 5. FESEM images (inset with schematic diagrams) of samples prepared at 160 °C for 16 h under different amount of SDS: (a) 5 mmol and (b) 15 mmol.

The products prepared under a determined amount of SDS (10 mmol) with different ratios of Sn^{4+} to L-cystine are shown in Fig. 6. As the stoichiometric ratio of SnS_2 is 1:2, more or less than the standard ratio of Sn^{4+} to L-cystine would not be advantageous for the formation of final products with well-defined hierarchical structure. Figure 6(a) shows the FESEM image of samples obtained under a relatively small molar ratio (1:4) of Sn^{4+} to L-cystine. In other words, the quantity of the S source is larger than the standard value for SnS_2. In Fig. 6(a), hierarchical bulk architectures can clearly be observed. As L-cystine is excessive, the high concentration of S^{2-} anions released from L-cystine will accelerate the crystal growth of SnS_2 under hydrothermal surroundings. In this condition, some of the SnS_2 architectures will not develop individually, but will combine mutually to form a bulk architecture. Interestingly, by further increasing the amount of L-cystine to a molar ratio (Sn^{4+}: L-cystine) of 1:10, architectures consisting of large nanosheets are fabricated, as shown in Fig. 6(b). An excess of L-cystine not only offers abundant S^{2-} anions for crystal growth, but also enhances the interaction between Sn^{4+} and some functional groups (e.g., –SH, –COOH, and $–NH_2$) in L-cystine molecules, which prevents coordination between Sn^{4+} and SDS. Moreover, the high density of negative charge in solution due to the large amount of S^{2-} anions will influence the formation of spherical SDS soft templates, which also carry negative charges. The aspects presented above contribute to the formation of large nanosheet architectures rather than spherical hierarchical nanostructures in the presence of excess L-cystine.

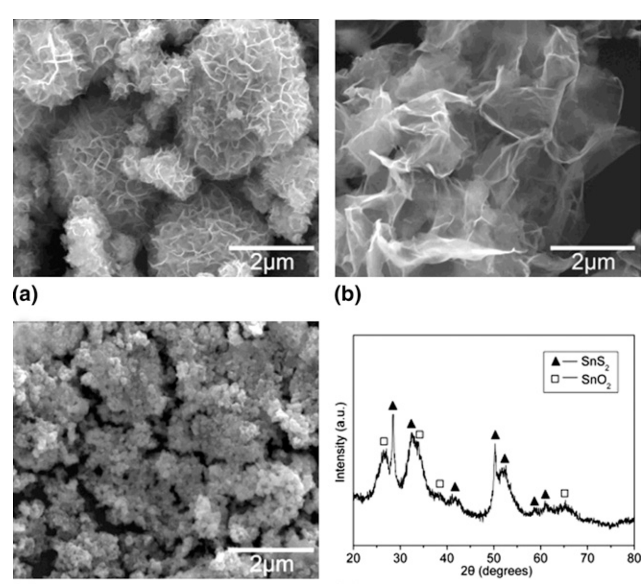


FIG. 6. FESEM images of products synthesized under a determined amount of SDS (10 mmol) with different ratios of Sn^{4+} to L-cystine: (a) 1:4, (b) 1:10, and (c) 1:1. (d) XRD pattern of the samples in part (c).

In the presence of excess Sn^{4+}, nanoparticles combined with some bulk structures are prepared, as shown in Fig. 6(c). In aqueous surroundings, excess Sn^{4+} hydrolyze to form $Sn(OH)_4$ and to produce SnO_2 through dehydration. This finding is confirmed by the XRD pattern, as shown in Fig. 6(d). From Fig. 6(d), diffraction peaks of both SnS_2 and SnO_2 phases can be observed, indicating the coexistence of hydrolysis of Sn^{4+} ions and reactions between L-cystine and Sn^{4+} ions.

C. Oxidizing conversion from SnS_2 to SnO_2 nanostructures

The as-assembled hierarchical SnS_2 hollow nanoarchitectures were further oxidized into SnO_2 for gas sensor application by annealing at 550 °C for 2 h in air. As shown in the low-magnification TEM image of products after annealing [Fig. 7(a)], the morphology of the hierarchical hollow nanostructure is similar to that of SnS_2, confirming the stability of the structure. This structure can be ascribed to the SDS-assisted assembly process and not the simple aggregation of nanoflakes. Interestingly, seen from the high-magnification TEM image [Fig. 7(b)], it is revealed that numerous pores distribute through the nanoflakes in a high density. The pore-size distribution ranges from 3 to 6 nm, which is in the region of mesopores.[32] The porous structure can also be confirmed by the HRTEM image [Fig. 7(c)]. The ring-like SAED pattern inserted in Fig. 7(c) suggests a polycrystalline structure. The presence of obvious discrete spots in the SAED pattern and clear lattice fringes in the HRTEM image indicate crystallinity as well. Seen from EDX spectrum [Fig. 7(d)], the oxidizing conversion is composed of Sn and O elements in a molar ratio (Sn:O) of 1:1.986 without the presence of S element, indicating the replacement of S by O atoms during the annealing process. As the radius of S atoms (0.109 nm) is larger than that of O atoms (0.065 nm), structural parameters such as bond lengths and angles are changed accordingly, resulting in a distortion of the crystal lattice which is similar to the reports on ZnO nanowires converted from ZnS.[33] In this condition, internal tensile stress in crystals is formed and will be released through the breakup of the crystal lattice, leading to the formation of the porous structure.

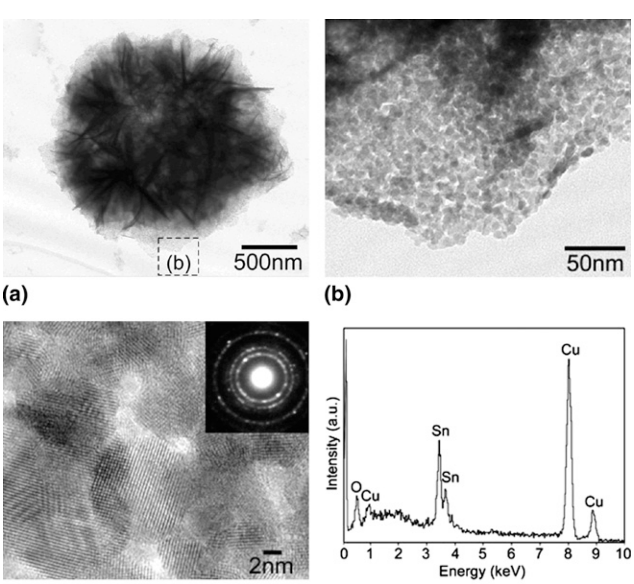


FIG. 7. (a) Low magnification and (b) high magnification TEM photographs, (c) HRTEM image (with an inserted corresponding SAED pattern), and (d) EDX spectrum of the products converted from SnS_2 by annealing at 550 °C for 2 h in air.

Figure 8 shows the N_2 adsorption/desorption isotherm and pore-size distribution of porous hierarchical SnO_2 hollow nanostructures. In Fig. 8, a type-IV isotherm with a hysteresis loop ranging from 0.4 to 1.0 (P/P_0) can be observed, which confirms the characteristic of mesoporous materials.[34] By BET analysis, it is found that the as-obtained SnO_2 products show a surface area of 136 m^2g^{-1}. The inserted pore-size distribution curve exhibits a bimodal shape with a narrow peak located at 3.9 nm and a broad one centered at 12.9 nm. The 3.9 nm peak is ascribed to the pores in the nanoflakes, which is in agreement with the TEM and HRTEM observations. On the other hand, as the hierarchical hollow nanostructures are composed of numerous nanoflakes, the interstices between adjacent nanoflakes can be thought to contribute to the formation of larger pores as revealed by the pore-size distribution curve. The special hierarchical SnO_2 nanoarchitectures with porous and hollow structures combined with a large surface area enable them possess a remarkably improved adsorption/desorption ability towards gases, indicating their potential applications in gas sensors with enhanced gas-sensing performance.

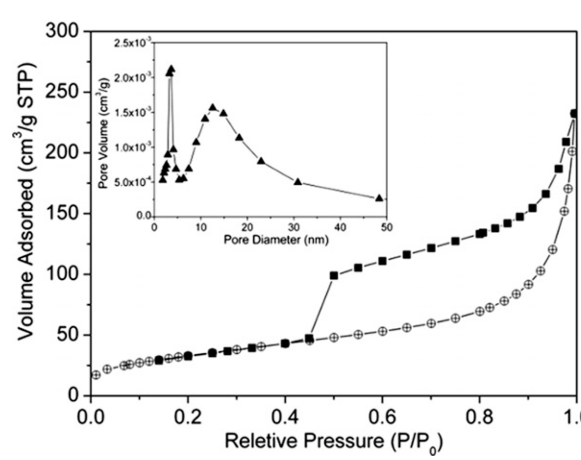


FIG. 8. Typical N_2 adsorption/desorption isotherm and pore-size distribution curve (inset) of the porous hierarchical SnO_2 hollow nanostructure.

D. Gas-sensing performance in drug precursor detection

As drug precursors can be poisonous and can potentially be used to synthesize drugs, their detection is significant for both human health and public safety. In our study, acetone and ethyl ether are typical drug precursors restricted in the UN Convention Against Illicit Traffic in Narcotic Drugs and Psychotropic Substances which was enacted in 1988. Another precursor in our study, chloroform, was listed as a drug precursor by the Ministry of Public Security of China in 2005. Figure 9 shows the gas-sensing performance of a gas sensor based on the as-prepared porous hierarchical SnO_2 hollow nanoarchitecture at a series of drug precursor concentrations ranging from 50 to 500 ppm. Seen from the real-time responses in gas detection [Fig. 9(a)], the current increases rapidly as the target gas is injected, and it decreases promptly while releasing target gases by the input of dry air, indicating a sensitive performance of the as-fabricated gas sensor.

The gas-sensing mechanism can be explained as follows. In ambient air, O_2 is adsorbed on the surface of SnO_2. It traps and reacts with electrons from SnO_2 to produce negative oxygen species such as O_2^-, O^-, and O^{2-}. As the electrons from sensing materials are trapped by adsorbed oxygen species, a space charge layer which serves as a potential barrier for electron transfer is formed on the surface of the SnO_2, resulting in a high air resistance.[35] This mechanism is similar to some other metal oxide semiconductors, such as In_2O_3 and ZnO.[36,37] When SnO_2 is exposed to acetone, chloroform, and ethyl ether, which are reducing gases, reactions between negative oxygen species and target gases occur, leading to the release of electrons and thereby a decrease in the resistance. Under a constant voltage, current thus increases following Ohm's law. Since the sensitivity of gas sensing depends on the surface area of the substrate, the hierarchical and hollow structures enable the SnO_2 nanomaterials to possess excellent gas-sensing properties. In addition, as seen in Fig. 9(a), the sensitivity increases at elevated concentrations of target gases, which can be confirmed by the calculated values as shown in Fig. 9(b). The results show that the as-fabricated gas sensor exhibits a high sensitivity toward drug precursors, indicating that it can potentially be applied in environmental monitoring and public safety investigation.

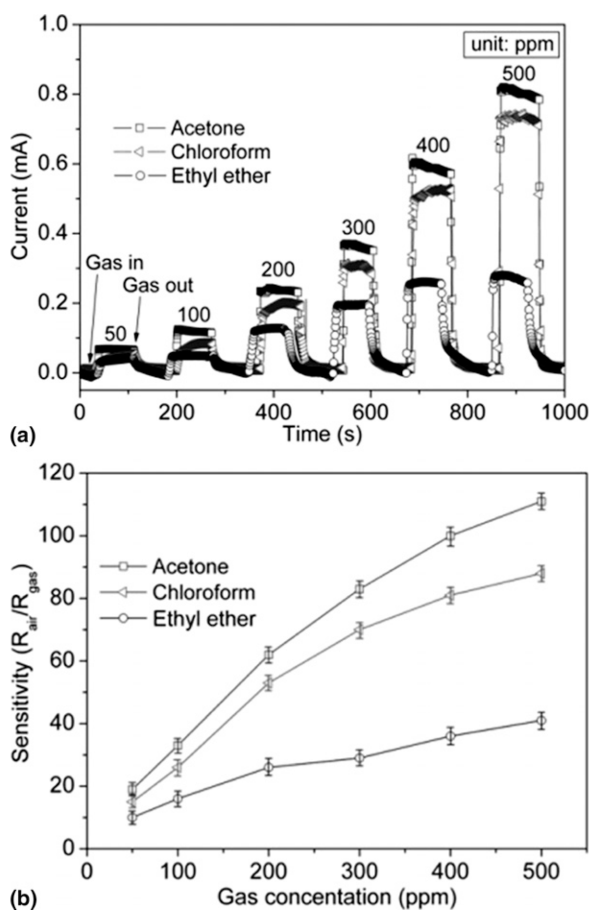


FIG. 9. Gas-sensing performance of the gas sensor based on porous hierarchical SnO_2 hollow nanoarchitecture toward drug precursors: (a) real-time responses and (b) corresponding sensitivities.

A comparison of gas-sensing performance between gas sensors based on assembled and unassembled nanoarchitecture was performed. For comparison, synthesis of SnS_2 structures without the assistance of SDS was carried out, followed by an annealing process to form SnO_2 conversion. The as-obtained SnO_2 materials exhibit a flower-like morphology with a few nanosheets grown on the surface as shown in the inset of Fig. 10(a), which is close to the samples displayed in Fig. 5(a). Under specified conditions, the gas-sensing abilities based on assembled and unassembled SnO_2 nanomaterials were measured. From the real-time responses to drug precursors [Figs. 10(a)–10(c)], it can be observed clearly that the assembled SnO_2 gas sensor shows remarkably enhanced gas sensing performance as compared to the unassembled SnO_2. The gas sensor based on the assembled SnO_2 nanoarchitecture exhibited faster response speed than the unassembled sample. This can be ascribed to the enhanced gas diffusion through the porous structure to the entire active surface of sensing materials.[38] According to the Knudsen diffusion equation,[39] the diffusion coefficient of gas is directly proportional to the porosity and pore diameter. As presented above, well-defined hierarchical SnO_2 nanomaterials were assembled by many radial nanoflakes. This means there is a large amount of nanopores dispersed all through the nanoflakes. In addition, there are many larger pores between adjacent nanoflakes that can be supported by the BET results (Fig. 8). Therefore, the diffusion coefficient of gas in the assembled SnO_2 nanostructures is larger than in the unassembled samples,

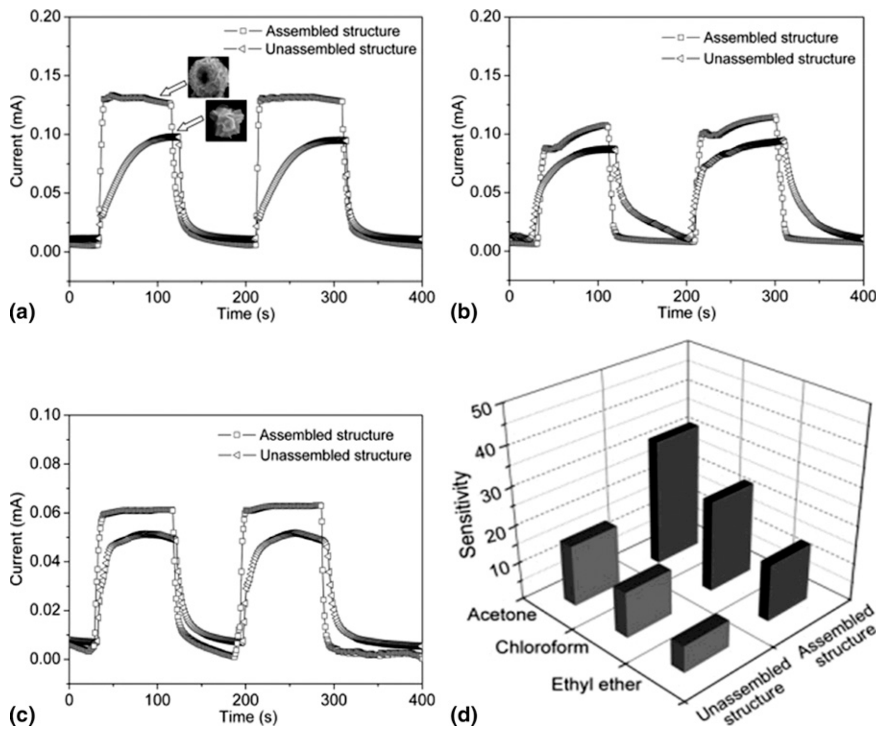


FIG. 10. Comparison of gas-sensing performance between gas sensors based on assembled and unassembled SnO_2 nanostructures toward drug precursors at 100 ppm: (a) acetone, (b) chloroform, and (c) ethyl ether. (d) The corresponding sensitivities.

leading to the decrease of gas response/recovery times which were defined as the time for the sensor to achieve 90% of total resistance change in the cases of gas injection or discharge. Moreover, as seen from the calculated sensitivities [Fig. 10(d)], it is found that the gas sensor based on assembled SnO_2 nanoarchitecture exhibits threefold sensitivity as compared to the unassembled samples. Both the porous hollow structure and the large surface area of the assembled hierarchical SnO_2 nanomaterial are advantageous for achieving an excellent adsorption/desorption ability for the target gases, which can be explanations for the significantly enhanced gas-sensing performance.

IV. CONCLUSIONS

In summary, hierarchical and hollow SnS_2 nanostructures were prepared via a surfactant-assisted assembly process. The as-synthesized SnS_2 nanomaterials as precursors were further oxidized to form porous SnO_2 conversion for the investigation of their gas-sensing performance toward drug precursors. The formation mechanism and role of SDS during the assembly process were demonstrated. It was revealed that the morphology and structure of the final products were greatly influenced by two aspects, the concentration of SDS, and the ratio of Sn to S in the source. The results of gas-sensing measurements showed that porous and hierarchical SnO_2 hollow nanostructures were sensitive to drug precursors, and exhibited remarkably improved gas-sensing properties compared with the unassembled SnO_2 nanomaterials with solid interior. These results point toward promising applications in environmental monitoring and public safety investigation. Moreover, it is expected that the SDS-assisted assembly process could potentially be developed as a general approach for assembling specific hierarchical nanoarchitectures with enhanced properties for applications.

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